Experiential Learning Project Assessment-I on "Inventory Stock Analysis System"

under the course

Python ProgrammingLanguage(N-PCCCM304P)

Presented By

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- GitHub account, repository creation, uploading code, meaningful commits, collaboration, individual contribution

Problem statement :

- Managing inventory manually is time-consuming and prone to errors
- It is difficult to keep track of stock after every sale.
- Low-stock products may not be identified on time.
- A simple Python-based system can automate stock tracking and analysis.

Objective:

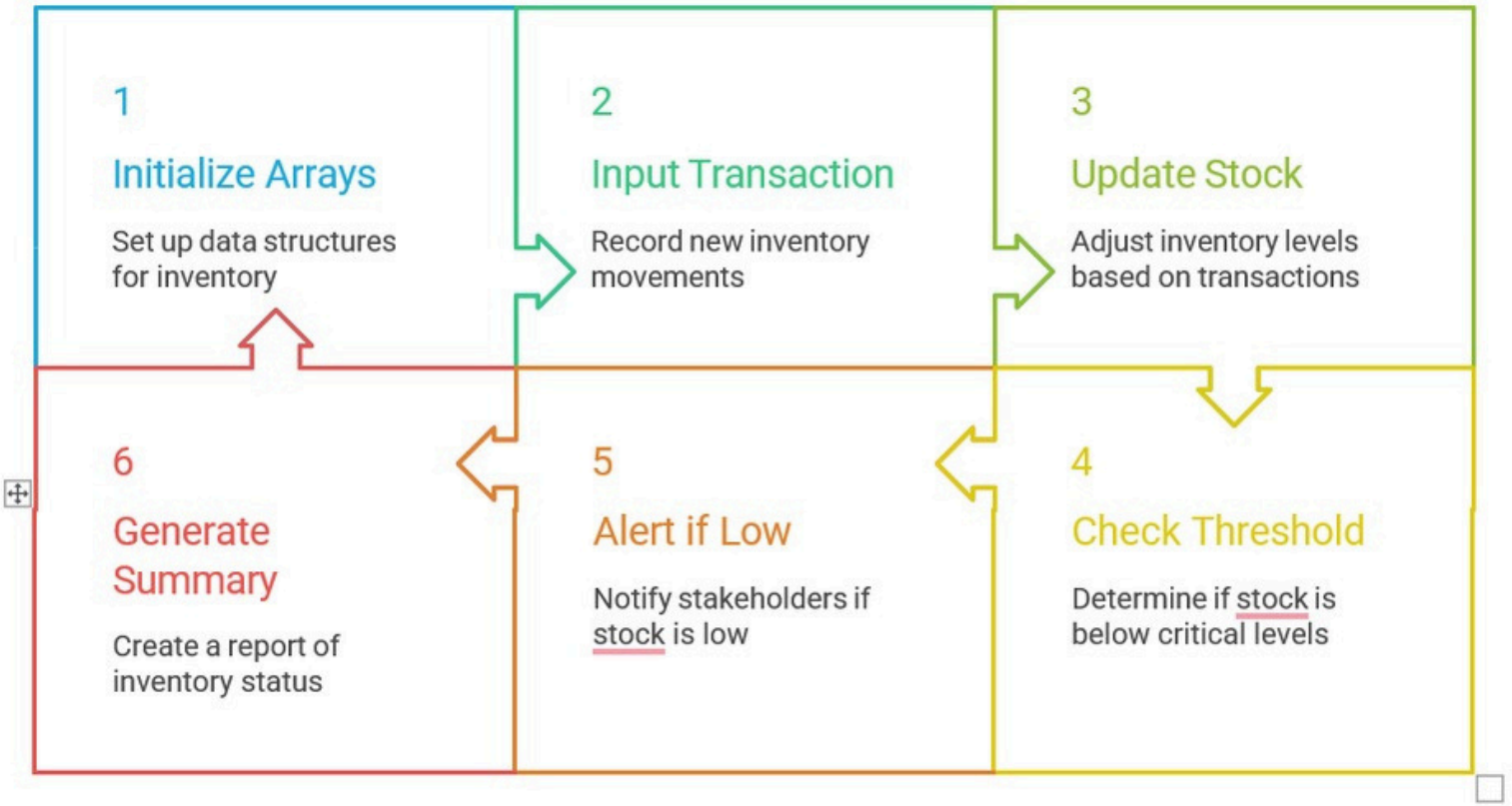
- To maintain product stock levels using Python lists.
- To record the quantity of products sold.
- To automatically update the remaining stock.
- To identify products that require restocking.
- To display an item-wise inventory summary

Algorithm:

Start the program.

- Store product names, stock quantities, and minimum stock levels.
- Display the available stock of each product.
- Enter the quantity sold.
- Subtract the sold quantity from the available stock.
- Check whether the remaining stock is below or equal to the minimum stock level.
- Display “RESTOCKING REQUIRED!” if the stock is low.
- Otherwise, display “Stock is Sufficient.”
- Display the final item-wise stock summary.
- Stop the program.

Flowchart


1
Initialize Arrays
Set up data structures for inventory
2
Input Transaction
Record new inventory movements
3
Update Stock
Adjust inventory levels based on transactions
4
Check Threshold
Determine if stock is below critical levels
5
Alert if Low
Notify stakeholders if stock is low
6
Generate Summary
Create a report of inventory status

Understanding python logic:

```python
# Inventory Stock Analysis System

products = ["Laptop", "Mouse", "Keyboard", "Monitor"]
stock = [10, 25, 15, 12]
minimum = [5, 10, 5, 5]

print("INVENTORY STOCK ANALYSIS SYSTEM")

for i in range(len(products)):
    print("\n", products[i], "- Stock:", stock[i])

    sold = int(input("Enter quantity sold: "))
    stock[i] -= sold

    if stock[i] <= minimum[i]:
        print("RESTOCKING REQUIRED!")
    else:
        print("Stock is Sufficient.")

print("\n--- FINAL STOCK SUMMARY ---")

for i in range(len(products)):
    print(products[i], ":", stock[i], "units")
```

lists → for loop → input → stock calculation → if-else → final summary.

GitHub account, repository creation, uploading code, meaningful commits, collaboration, individual contribution:

1)Create a GitHub account and use it to store and manage the Inventory project code online.

2)Createarepository namedInventory-Management-Systemand add the project files/code to it.

3)UploadthePython sourcecodeand otherrequiredproject files to the repository.

4)Meaningful Commits:Fixed stock validation

5)If working in a team, members can clone the repository, create branches, make changes, and push their work to GitHub.

6)https://github.com/shreyagml25-maker/invontory-stock-analysis.git

Expected Output/ Result:

```
==========================================
       INVENTORY STOCK ANALYSIS SYSTEM
==========================================

--------- CURRENT INVENTORY ---------
Laptop : 10 units
Mouse : 25 units
Keyboard : 15 units
Monitor : 12 units
Headphones : 8 units

--------- SALES ENTRY ---------

Product: Laptop
Available Stock: 10
Enter quantity sold: 6
Remaining Stock: 4
RESTOCKING REQUIRED!

Product: Mouse
Available Stock: 25
Enter quantity sold: 10
Remaining Stock: 15
Stock is Sufficient.
```

```
Product: Monitor
Available Stock: 12
Enter quantity sold: 8
Remaining Stock: 4
RESTOCKING REQUIRED!

Product: Headphones
Available Stock: 8
Enter quantity sold: 4
Remaining Stock: 4
RESTOCKING REQUIRED!


==========================================
           FINAL STOCK SUMMARY
==========================================
Laptop : 4 units - RESTOCK
Mouse : 15 units - AVAILABLE
Keyboard : 10 units - AVAILABLE
Monitor : 4 units - RESTOCK
Headphones : 4 units - RESTOCK
==========================================
       INVENTORY ANALYSIS COMPLETE
==========================================
```

PHASE-II

1) Inventory Management & Processing

- Product Inventory Maintenance – Stores product names and available stock levels.
- Sales Entry –Recordsthe quantity ofproductssold.
- Stock Validation – Checks whether the requested quantity is available.
- Automatic Stock Update – Deducts the sold quantity from the available inventory.
- Stock Level Monitoring – Continuously tracks the remaining quantity of each product.
- Inventory Status Update – Displays the current stock status after processing

2) Stock Analysis

- Stock Level Monitoring – Tracks the remaining quantity of each product.
- Minimum Stock Comparison – Compares available stock with the predefined minimum level.
- Low-Stock Identification – Automatically identifies products requiring restocking.
- Restocking Alerts – Displays an alert when stock reaches or falls below the minimum level.
- Stock Status Classification – Shows whether the stock is sufficient or requires restocking.

3) Data Validation & Error Handling

- Stock Validation – Prevents sales exceeding available inventory.
- Input Validation – Ensures the entered quantity can be processed correctly.
- Error Prevention – Avoids incorrect stock calculations.
- Inventory Accuracy – Maintains reliable stock information after each transaction.
- Exception Handling – Provides an appropriate message when the requested quantity is unavailable.

PHASE III

AIM:-

To develop a Python-based Inventory Stock Analysis System that manages product details, updates stock quantities, identifies low-stock products, and calculates the total inventory value.

OBJECTIVES:-

To store and manage product names, quantities, and prices.

To add new products and update existing stock.

To identify products with low stock levels.

To calculate total stock quantity and inventory value.

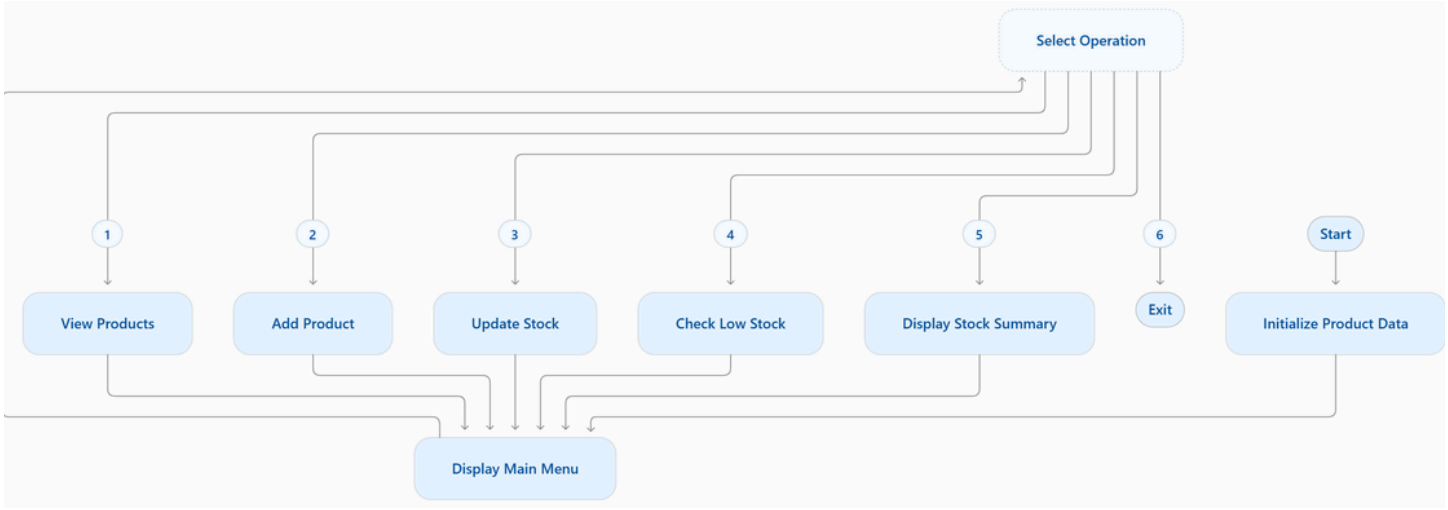
To generate a summary of available and low-stock products.

To reduce manual work and improve inventory management.

Algorithm:-

1. Start the program.
2. Initialize product names, stock quantities, minimum stock levels,and prices.
3. Display the main menu.
4. Allow the user to view products, add a product, update stock, check low-stock alerts, or view the summary.
5. Perform the selected operation.
6. Display the updated inventory information.
7. Repeat until the user selects Exit.
8. Stop the program.

FLOWCHART:-


Select Operation
1
2
3
4
5
6
Start
View Products
Add Product
Update Stock
Check Low Stock
Display Stock Summary
Exit
Initialize Product Data
Display Main Menu

CODE:-

```
# Inventory Stock Analysis System - Phase 3

products = ["Laptop", "Mouse", "Keyboard", "Monitor", "Headphones"]
stock = [10, 25, 15, 12, 8]
minimum = [5, 10, 5, 5, 5]
price = [50000, 500, 1200, 8000, 1500]

while True:
print("\n--- INVENTORY STOCK ANALYSIS ---")
print("1. View Products")
print("2. Add Product")
print("3. Update Stock")
print("4. Low Stock Alert")
print("5. Stock Summary")
print("6. Exit")

choice = input("Enter your choice: ")

if choice == "1":
print("\nProduct\t\tStock\tPrice")
for i in range(len(products)):
print(products[i], "\t", stock[i], "\t", price[i])

elif choice == "2":
name = input("Enter product name: ")

if name.lower() in [p.lower() for p in products]:
print("Product already exists!")
else:
try:
qty = int(input("Enter quantity: "))
cost = float(input("Enter price: "))
min_qty = int(input("Enter minimum stock: "))

if qty < 0 or cost < 0 or min_qty < 0:
print("Values cannot be negative!")
else:
products.append(name)
stock.append(qty)
price.append(cost)
minimum.append(min_qty)
print("Product added successfully!")
except ValueError:
print("Please enter valid numbers.")

elif choice == "3":
name = input("Enter product name: ")
found = False

for i in range(len(products)):
if products[i].lower() == name.lower():
try:
qty = int(input("Enter new stock quantity: "))
if qty >= 0:
stock[i] = qty
print("Stock updated successfully!")
else:
print("Quantity cannot be negative.")
except ValueError:
print("Please enter a valid quantity.")
found = True
break

if not found:
print("Product not found!")

elif choice == "4":
print("\nLow Stock Products:")
found = False

for i in range(len(products)):
if stock[i] <= minimum[i]:
print(products[i], "- Stock:", stock[i])
found = True

if not found:
print("No low-stock products.")

elif choice == "5":
total_products = len(products)
total_stock = sum(stock)
total_value = 0

for i in range(len(products)):
total_value += stock[i] * price[i]

print("\n--- STOCK SUMMARY ---")
print("Total Products:", total_products)
print("Total Stock Units:", total_stock)
print("Total Inventory Value: Rs.", total_value)

elif choice == "6":
print("Invalid choice. Please try again.")
```

OUTPUT:

--- INVENTORY STOCK ANALYSIS ---
1. View Products
2. Add Product
3. Update Stock
4. Low Stock Alert
5. Stock Summary
6. Exit

Enter your choice: 5

--- STOCK SUMMARY ---
Total Products: 5
Total Stock Units: 70
Total Inventory Value: Rs. 660000.0

Final Conclusion

The Inventory Stock Analysis System – Phase 3 is a Python-based project developed to simplify inventory management. It allows users to view products, add new products, update stock quantities, identify low-stock items, and calculate the total inventory value. The system reduces manual effort, improves stock accuracy, and helps users manage product information efficiently. Overall, this project provides a simple and effective solution for inventory stock analysis and can be enhanced in the future with database integration and advanced reporting features.

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Thank You